

What is a Function?

☞ A **function** is a block of code used to perform a task.

→ □ It runs only when we **call** it

Why Functions are Important?

- Reuse code
- Avoid repetition
- Makes code clean
- Easy to debug

◆ Syntax of Function

```
function functionName() {  
    // code  
}
```

◆ Example 1 (Basic Function)

```
function sayHello() {  
    console.log("Hello");  
}
```

☞ Calling function:

```
sayHello();
```

✦ Output:

```
Hello
```

◆ Function with Parameters

☞ Parameters = input values

```
function add(a, b) {  
    console.log(a + b);  
}
```

☞ Calling:

```
add(2, 3);
```

✈ Output:

```
5
```

◆ Function with Return

☞ Returns value

```
function multiply(a, b) {  
    return a * b;  
}
```

```
let result = multiply(2, 3);  
console.log(result);
```

✈ Output:

```
6
```

What is `alert` in JavaScript?

🔗 `alert` is used to **show a popup message (dialog box)** on the screen.

◆ Syntax

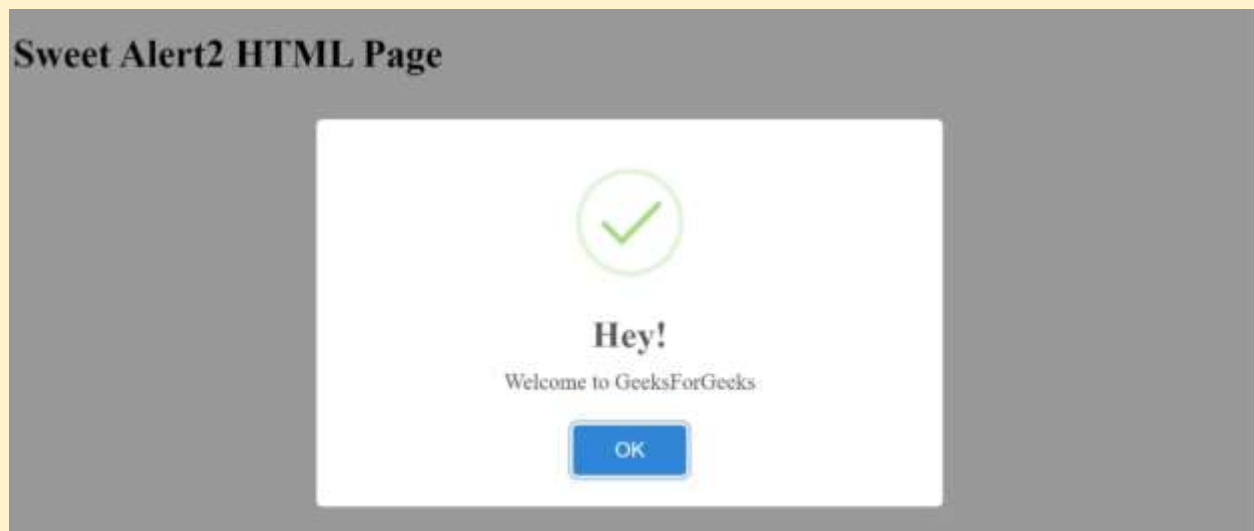
```
alert("message");
```

◆ Example

```
alert("Hello");
```

✦ Output:

A popup box appears with **Hello**



◆ What is an Event?

☞ **Event = action performed by user or browser**

✦ Examples:

- Clicking a button
- Typing in input
- Moving mouse
- Page loading

→ ☐ When event happens → JavaScript runs a function

◆ Basic Syntax

```
<button onclick="test()">Click</button>
```

```
<script>
function test() {
    alert("Clicked");
}
</script>
```

✦ Flow:

- User clicks → `onclick` event
 - Event calls function → function runs
-

◆ Most Important Events

✓ 1. `onclick`

☞ When user clicks

```
<button onclick="clickMe()">Click</button>

<script>
function clickMe() {
    alert("Clicked");
}
</script>
```

✓ 2. oninput

☞ When user types (real-time)

```
<input oninput="typing()">

<script>
function typing() {
    console.log("Typing...");
}
</script>
```

◆ Where will "Typing..." appear?

☞ It will appear in the **browser console**, NOT on the webpage.

◆ What is console.log() ?

☞ console.log() is used to **print messages in the browser console**

◆ How to see it?

☞ Steps:

1. Open your webpage
2. Press **F12** (or Right click → Inspect)
3. Go to **Console tab**

☞ Now type in the input box

→ □ You will see:

Typing...
Typing...
Typing...

◆ Why not visible on screen?

☞ Because:

- `console.log()` → for developers (debugging)
 - It does NOT display on webpage
-

✓ 3. `onchange`

☞ When value changes (after leaving input)

```
<input onchange="changed()" ">
```

```
<script>  
function changed() {  
    alert("Changed");  
}  
</script>
```

✓ 4. `onmouseover`

☞ Mouse enters

```
<p onmouseover="hover()" ">Hover me</p>
```

```
<script>  
function hover() {  
    console.log("Mouse entered");  
}  
</script>
```

✓ 5. `onmouseout`

☞ Mouse leaves

```
<p onmouseout="out()" ">Move out</p>
```

```
<script>
```

```
function out() {  
    console.log("Mouse left");  
}  
</script>
```

✓ 6. onkeydown

☞ Key pressed

```
<input onkeydown="press()">  
  
<script>  
function press() {  
    console.log("Key pressed");  
}  
</script>
```

✓ 7. onsubmit

☞ Form submit

```
<form onsubmit="submitForm()">  
    <button type="submit">Submit</button>  
</form>  
  
<script>  
function submitForm() {  
    alert("Form submitted");  
}  
</script>
```

✓ 8. onload

☞ Page loaded

```
<body onload="start()">  
  
<script>  
function start() {  
    alert("Page loaded");  
}  
</script>
```

document.getElementById() and innerText

◆ What is document.getElementById() ?

☞ It is used to **select an HTML element using its id**

☞ **Syntax:**

```
document.getElementById("idName")
```

★ **Example:**

```
<p id="text">Hello</p>
document.getElementById("text");
```

☞ This selects the <p> element

◆ What is innerText ?

☞ It is used to **change or display text inside an element**

☞ **Syntax:**

```
element.innerText = "New Text";
```

◆ Combined Example

```
<p id="text">Hello</p>

<button onclick="changeText()">Click</button>

<script>
function changeText() {
    document.getElementById("text").innerText = "Changed!";
}
</script>
```


TASKS

1) Increment / Decrement / Reset Counter

```
<!DOCTYPE html>
<html>
<head>
  <title>Counter</title>
</head>
<body>

  <!-- DISPLAY VALUE -->
  <h2 id="count">0</h2>

  <!-- BUTTONS -->
  <button onclick="increment()">Increment</button>
  <button onclick="decrement()">Decrement</button>
  <button onclick="reset()">Reset</button>

  <script>
    // STORE VALUE
    let count = 0;

    // INCREMENT FUNCTION
    function increment() {
      count++;
      document.getElementById("count").innerText = count;
    }

    // DECREMENT FUNCTION
    function decrement() {
      count--;
      document.getElementById("count").innerText = count;
    }

    // RESET FUNCTION
    function reset() {
      count = 0;
      document.getElementById("count").innerText = count;
    }
  </script>

</body>
</html>
```

0

Increment

Decrement

Reset

Task -2

2) Simple Calculator

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Calculator</title>
</head>
<body>

  <!-- INPUT BOXES -->
  <input type="number" id="num1" placeholder="Enter first number">
  <input type="number" id="num2" placeholder="Enter second number">

  <br><br>

  <!-- BUTTONS -->
  <button onclick="add()">Add</button>
  <button onclick="sub()">Subtract</button>
  <button onclick="mul()">Multiply</button>
  <button onclick="div()">Divide</button>

  <!-- RESULT -->
  <h2 id="result">Result: 0</h2>

  <script>
    function add() {
      let a = Number(document.getElementById("num1").value);
      let b = Number(document.getElementById("num2").value);
```

```
        document.getElementById("result").innerText = "Result: " + (a + b);
    }

    function sub() {
        let a = Number(document.getElementById("num1").value);
        let b = Number(document.getElementById("num2").value);
        document.getElementById("result").innerText = "Result: " + (a - b);
    }

    function mul() {
        let a = Number(document.getElementById("num1").value);
        let b = Number(document.getElementById("num2").value);
        document.getElementById("result").innerText = "Result: " + (a * b);
    }

    function div() {
        let a = Number(document.getElementById("num1").value);
        let b = Number(document.getElementById("num2").value);
        document.getElementById("result").innerText = "Result: " + (a / b);
    }
}
</script>
</body>
</html>
```

Result: 4

Task – 3

Todo List

```
<!DOCTYPE html>
<html>
<head>
  <title>To-Do List</title>
</head>
<body>

  <h2>To-Do List</h2>

  <!-- INPUT -->
  <input type="text" id="task" placeholder="Enter task">

  <!-- ADD BUTTON -->
  <button onclick="addTask()">Add</button>

  <!-- LIST -->
  <ul id="list"></ul>

  <script>
    function addTask() {
      // GET INPUT VALUE
      let task = document.getElementById("task").value;

      // CREATE NEW LIST ITEM
      let li = document.createElement("li");

      // ADD TEXT TO LIST ITEM
      li.innerText = task;

      // ADD TO LIST
      document.getElementById("list").appendChild(li);

      // CLEAR INPUT
      document.getElementById("task").value = "";
    }
  </script>

</body>
</html>
```

To-Do List

- hii Hello
- nice to meet you

Task -4

Form Validation

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
</head>
<body>

  <h2>Form Validation</h2>

  <!-- FORM -->
  <form onsubmit="return validateForm()">

    Name: <input type="text" id="name"><br><br>
    Email: <input type="text" id="email"><br><br>
    Password: <input type="password" id="password"><br><br>
```

```

        <button type="submit">Submit</button>
    </form>

    <!-- MESSAGE -->
    <p id="error" style="color:red;"></p>

    <script>
        function validateForm() {

            let name = document.getElementById("name").value;
            let email = document.getElementById("email").value;
            let password = document.getElementById("password").value;

            // CHECK EMPTY
            if (name === "" || email === "" || password === "") {
                document.getElementById("error").innerText = "All fields are
required";
                return false;
            }

            // PASSWORD LENGTH CHECK
            if (password.length < 6) {
                document.getElementById("error").innerText = "Password must be at
least 6 characters";
                return false;
            }

            // SUCCESS
            alert("Form Submitted Successfully");
            return true;
        }
    </script>

</body>
</html>

```

Form Validation

Name:

Email:

Password:

127.0.0.1:5500 says

Form Submitted Successfully

OK

Task – 5

Toggle BG

```
<!DOCTYPE html>
<html>
<head>
  <title>Light Dark Mode</title>

  <style>
    /* LIGHT MODE (default) */
    body {
      background-color: white;
      color: black;
      text-align: center;
      margin-top: 50px;
    }

    /* DARK MODE */
    .dark {
      background-color: black;
      color: white;
    }
  </style>
</head>
<body>

  <h2>Light / Dark Mode</h2>

  <!-- BUTTON -->
  <button onclick="toggleMode()">Toggle Mode</button>

  <script>
    function toggleMode() {
      document.body.classList.toggle("dark");
    }
  </script>

</body>
</html>
```

Light / Dark Mode

Toggle Mode

Light / Dark Mode

Toggle Mode

Notes :

1. `document.body`

☞ Refers to the **entire** `<body>` of HTML page

```
document.body.style.background = "black";
```


✦ Used to:

- Change full page style
 - Apply class to whole page
-

◆ 2. `.classList.toggle()`

🔗 Used to **add/remove a CSS class**

```
document.body.classList.toggle("dark");
```

✦ Meaning:

- If class exists → remove
 - If not → add
-

◆ 3. `.innerText`

🔗 Used to **change text inside element**

```
document.getElementById("text").innerText = "Hello";
```

✦ Output shown on webpage

◆ 4. `.createElement()`

🔗 Creates a **new HTML element**

```
let li = document.createElement("li");
```

✦ It creates but does NOT show yet

◆ 5. `.appendChild()`

☞ Adds element to HTML

```
document.getElementById("list").appendChild(li);
```

✦ Now element appears on screen

◆ 6. `.value`

☞ Gets value from input field

```
let x = document.getElementById("num1").value;
```

✦ Always returns **string**

◆ 7. `Number()`

☞ Converts string → number

```
Number("5")    // 5
```

◆ 8. Combined (Important)

```
Number(document.getElementById("num2").value)
```

☞ Step by step:

1. Get input value → "5" (string)
2. Convert → 5 (number)

✦ Needed for calculations

🔥 Difference Table (Very Important)

Concept	Purpose	Example
<code>document.body</code>	whole page	change background
<code>classList.toggle()</code>	add/remove class	dark mode
<code>innerText</code>	change text	display result
<code>createElement()</code>	create element	<code></code>
<code>appendChild()</code>	add element	show <code></code>
<code>.value</code>	get input value	"5"
<code>Number()</code>	convert to number	5

◆ Simple Flow Example

```
let value = Number(document.getElementById("num").value);
```

🔗 Flow:

- `.value` → gets string
 - `Number()` → converts
 - Now usable for math
-

🔑 Easy Memory Trick

- **create** → **append** → **show**
 - **value** → **string**
 - **Number** → **convert**
 - **innerText** → **display**
 - **toggle** → **switch**
-

🔥 Final Understanding

🔗 These are used together:

- Input → `.value`

- Convert → `Number()`
- Logic → `function`
- Display → `innerText`
- UI change → `classList.toggle()`